

# ZIYI ZHU

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## EXPERIENCE

### Slingshot AI

*Member of Technical Staff*

New York, United States

Nov. 2024 – Present

- Led full end-to-end development of a foundation model on psychology, including **continual pre-training** on human therapy transcripts, **supervised fine-tuning** on golden conversations curated by clinical teams, and **alignment** on annotator-labeled preference datasets via offline and online policy optimization algorithms such as DPO and GRPO.
- Engineered scalable training infrastructure with hands-on experience in data preparation, cleaning, and multi-GPU distributed training with TRL and verl; scaled single-node training to **16k context** on **122B-parameter** models using PEFT techniques.
- Deployed language models and reward models (token and sequence classification) with vLLM and TensorRT with autoscaling in a managed Kubernetes cluster, serving both production traffic and development workloads.
- Developed value models to predict long-term outcomes and user behavioral signals, applying multi-objective RL optimization to **improve D1–D7 retention by ~3%** while increasing positive behaviors and therapeutic-process markers in production.
- Developed novel training methods, including synthetic preference datasets selected by learned value functions, and adversarial training for realistic user simulation, enabling offline evaluation and multi-turn policy optimization.
- Led design and development of an agentic prompt optimization system in which agents perform diagnostic code execution and propose targeted prompt changes, driving a **14% D1 retention improvement** over the quarter.
- Led initiatives to improve voice turn-taking experience; trained an audio-based endpointing model built on Whisper to detect end of turn, **reducing interruption rates by 70%** at equivalent latency targets.
- Developed internal tooling for annotation workflows and benchmark creation using TypeScript and Kotlin.

### Unlikely AI

*Senior Applied Scientist*

London, United Kingdom

Jan. 2023 – Nov. 2024

- Architected core components of a neurosymbolic AI platform combining LLMs with symbolic reasoning, prioritizing accuracy, trustworthiness, and explainability for high-stakes decision-making in regulated industries.
- Engineered semantic parsing and entity resolution models, achieving **50% improvement** over open-source alternatives through transformer fine-tuning on synthetic datasets for insurance and financial services applications.
- Redesigned the symbolic language by integrating LLMs with symbolic solvers, delivering **99% precision** on tasks with **90% cost reduction**; this technology formed the foundation of the platform sold to Lloyds Banking Group.
- Achieved significant reduction in reasoning complexity through optimization of ontological reasoning using imitation learning and reinforcement learning techniques, enabling real-time inference at scale.

### Chai Research

*Founding AI Researcher*

Palo Alto, United States

Dec. 2021 – Nov. 2022

- Built a social AI platform from inception to achieving **200k+ daily active users** and **1M+ daily messages**, establishing one of the largest real-world LLM deployments at the time.
- Achieved **60% D1 retention** through LLM fine-tuning and prompt engineering on curated conversational datasets, along with user-centric research and implementation of **20+ app features** in Flutter.
- Implemented encoder-based reward models trained on human preference data using rejection sampling, which **doubled user engagement** by increasing median conversation length by 100%.
- Designed, built, and deployed an offline reinforcement learning pipeline using PPO to maximize user engagement, marking one of the earliest successful applications of RLHF to social AI products at scale.
- Orchestrated the large-scale deployment of LLMs across CoreWeave GPU clusters, building high-throughput inference infrastructure that successfully served hundreds of models on thousands of GPUs with high availability.

### Seamless Capital

*Quantitative Researcher*

Cambridge, United Kingdom

Aug. 2021 – Jan. 2022

- Developed machine learning models (GBDT, MLP, transformers, and ensemble methods) for multi-market horse racing prediction, including Win/Place/Quinella, deployed in live automated trading on Betfair exchange.
- Engineered **50+ alpha signals** and specialized features through historical data analysis, time-series feature extraction, and odds-implied probability calibration, resulting in **10% P&L improvement**.
- Implemented back-testing infrastructure with cross-validation, walk-forward optimization, and Monte Carlo simulation for risk analysis, validated through Sharpe ratio, drawdown metrics, and live market execution performance.

### UBS Group AG

*Financing Quant Intern*

London, United Kingdom

June 2021 – Aug. 2021

- Optimized Stock Borrow Loan (SBL) desk operations through automated trade booking, position reconciliation, and settlement.
- Developed analytical tools on the Advanced Cloud Quantitative Analytics (ACQA) platform for repo rate curve construction, collateral valuation, and securities pricing models.

## PUBLICATIONS

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- **Z. Zhu**, D. R. Cahn, T. D. Hull, C. A. Stamatis, O. Tieleman, G. B. Freire, J. Chen. “Optimizing Sparse Outcomes Through Dense Behavioral Signals via Value-Guided Preference Distillation.” *Preprint*, 2026. [[arXiv:2609.14648](#)]
- **Z. Zhu**, O. Tieleman, A. Bukhtiyarov, J. Chen. “CyclicJudge: Mitigating Judge Bias Efficiently in LLM-based Evaluation.” *EMNLP 2026 (Main Conference)*. [[arXiv:2603.01865](#)]
- **Z. Zhu**, O. Tieleman, C. A. Stamatis, L. Smyth, T. D. Hull, D. R. Cahn, J. Chen, M. Malgaroli. “DIAL: Direct Iterative Adversarial Learning for Realistic Multi-Turn Dialogue Simulation.” *Findings of EMNLP 2026*. [[arXiv:2512.20773](#)]
- **Z. Zhu**, L. Smyth, S. Shinoda, J. Chen. “SAGE: Stochastic Prompt Optimization via Agent-Guided Exploration.” *EMNLP 2026 (Industry Track)*. [[arXiv:2606.18902](#)]
- R. Irvine, D. Boubert, V. Raina, A. Liusie, **Z. Zhu**, et al. “Rewarding Chatbots for Real-World Engagement with Millions of Users.” *Preprint*, 2023. [[arXiv:2303.06135](#)]

## EDUCATION

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**University of Cambridge, Christ’s College**

Cambridge, United Kingdom

*Master of Arts (Cantab.)*

May 2025

*Bachelor of Arts, Engineering*

Oct. 2018 – June 2021

- Results: First Class Honours, ranking in the **top 4% (12th out of 288 students)**, securing the Christ’s College Scholarship for **three consecutive years**.
- Modules: Information and Computer Engineering, Electrical and Information Sciences, and Instrumentation and Control.
- Achievements: Christ’s College Art Prize 2021, CUES Hackathon Champion 2021, VOMG Most Promising Musician 2017, and dual Silver Awards in Singapore Physics Olympiad (2014, 2016).

## ACTIVITIES

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**Smartathon (Smart City Challenge) by Saudi Data & AI Authority (SDAIA)**

Riyadh, Saudi Arabia

*Second Prize Winner*

Apr. 2023

- Secured **2nd place** among **14,731** teams globally, earning **SAR 97,500** in prize money and presenting the solution at Riyadh.
- Developed a pothole severity classification system combining YOLOv5 for real-time 2D object detection and DiffNet for monocular depth estimation and 3D surface reconstruction, enabling automated road maintenance optimization.

*Personal Projects*

- Engineered and deployed a profitable tennis match prediction system on Betfair, consistently achieving **~5%** daily returns.
- Derived closed-form probability model for stop-limit order execution using Optional Stopping Theorem and exponential martingales on geometric Brownian motion, validated through Monte Carlo simulations, achieving **<1%** relative error.

## SKILLS & INTERESTS

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**Languages:** Chinese (Native), English (Fluent), Japanese (Beginner)

**Programming:** Python, TypeScript, JavaScript, Kotlin, Java, C++, SQL, Flutter, React, MATLAB

**LLM Training & Inference:** PyTorch, Hugging Face Transformers, TRL, verl, DeepSpeed, vLLM, TensorRT, PEFT techniques (LoRA, QLoRA)

**Domains:** Natural Language Processing, Computer Vision, Diffusion Models, Audio Processing (speech endpointing & generation)

**Classical ML & Vision:** XGBoost, LightGBM, OpenCV, scikit-learn, image segmentation, depth estimation

**Infrastructure:** Distributed training (multi-node, multi-GPU), Docker, Kubernetes, GCP, AWS, Weights & Biases, Nebius Cloud

**Hobbies:** Percussive fingerstyle guitar, piano, drum set, visual art, oil painting, photography, pool, tennis, table tennis, and travel